

91262



NEW ZEALAND QUALIFICATIONS AUTHORITY
MANA TOHU MĀTAURANGA O AOTEAROA

QUALIFY FOR THE FUTURE WORLD
KIA NOHO TAKATŪ KI TŌ ĀMUA AOI

2

DRD

SUPERVISOR'S USE ONLY

worked
solutions
thinking
grades

Level 2 Mathematics and Statistics, 2015

91262 Apply calculus methods in solving problems

2.00 p.m. Tuesday 10 November 2015
Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Apply calculus methods in solving problems.	Apply calculus methods, using relational thinking, in solving problems.	Apply calculus methods, using extended abstract thinking, in solving problems.

Check that the National Student Number (NSN) on your admission slip is the same as the number at the top of this page.

You should attempt ALL the questions in this booklet.

Make sure that you have Resource Sheet L2-MATHF.

Show ALL working.

If you need more space for any answer, use the page(s) provided at the back of this booklet and clearly number the question.

Check that this booklet has pages 2–12 in the correct order and that none of these pages is blank.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

TOTAL

ASSESSOR'S USE ONLY

QUESTION ONE

- (a) A function f is given by $f(x) = x^4 + 2x^2 - 5$

Find the gradient of the graph of the function at the point where $x = -1$.

$$f'(x) = 4x^3 + 4x$$

$$f'(-1) = 4(-1)^3 + 4(-1) = -8$$

gradient
→ diff

(a)

- (b) $f(x) = 8 + 3x + x^2 - \frac{x^3}{3}$

For what values of x is f a decreasing function?

Justify your answer.

You must show the use of calculus.

$$f'(x) = 3 + 2x - x^2$$

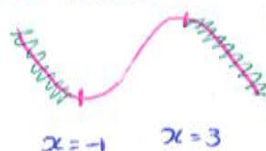
$$3 + 2x - x^2 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

$$x = -1 \text{ or } x = 3$$

• negative cubic



• decreasing

• need stationary pts to find where decreasing

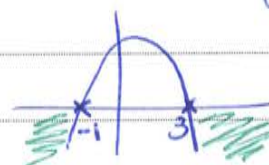
• stat pts → $f'(x) = 0$

justification (as well as pic above)

• f will be a decreasing function when gradient is negative.

• $f'(x)$ is a negative parabola

so function is decreasing when $x < -1$ and $x > 3$.



negative (below zero) here

ⓐ = regions identified + justification.

- (c) Calculate the rate at which the volume of a cube is changing with respect to its length, at the instant when the length of each edge of the cube is 5 cm.

$$V = l^3$$



$$\frac{dV}{dl} = 3l^2$$

when $l = 5$ $\frac{dV}{dl} = 3(5)^2 = 75 \text{ cm}^3/\text{cm}$ (m)

- (d) A train passes a signal at a velocity of 40 m s^{-1} .
The train's acceleration, $a \text{ m s}^{-2}$, t seconds after it passes the signal, can be modelled by the function

$$a(t) = (16 - 2t)$$

- (i) What is the greatest speed attained by the train after it passes the signal?

• want max velocity; at $t=0$, $v=40 \text{ m/s}$;

$$v(t) = 16t - t^2 + C$$

$$40 = 16(0) - (0)^2 + C$$

$$C = 40$$

$$v(t) = 16t - t^2 + 40$$

max velocity occurs

$$\rightarrow \text{when } v'(t) = a(t) = 0$$

$$16 - 2t = 0$$

$$16 = 2t$$

$$t = 8$$

$$\text{when } t=8 \quad v(t) = 16(8) - 8^2 + 40 = 104 \text{ m/s}$$

max speed at $t=8$ is 104 m/s

- (ii) How far past the signal does the train travel before it stops?

• stops when $v(t) = 0$

• $t=0$, $d=0$, $v=40$

$$v(t) = 16t - t^2 + 40 = 0$$

$$t = 18.1980 \quad \text{or} \quad t = -2.1980$$

*cannot have negative time as this will be prior to passing signal.

train stops after 18.1980 seconds.

$$s(t) = 8t^2 - \frac{t^3}{3} + 40t + C$$

$$0 = 8(0)^2 - \frac{(0)^3}{3} + 40(0) + C$$

$$C = 0$$

$$s(t) = 8t^2 - \frac{t^3}{3} + 40t$$

when $t = 18.1980 \text{ sec}$

$$s(18.1980) = 8(18.1980)^2 - \frac{(18.1980)^3}{3} + 40(18.1980)$$

distance = 1368 m
till stopped.

QUESTION TWO

- (a) The gradient of function f is given by $f'(x) = 4x - 3$.
The point $(4, 6)$ lies on the graph of the function.

Find the equation of the function f .

$$f(x) = 2x^2 - 3x + c$$

sub $(4, 6)$ to find c

$$6 = 2(4)^2 - 3(4) + c$$

$$6 = 32 - 12 + c$$

$$6 = 20 + c$$

$$c = -14$$

$$f(x) = 2x^2 - 3x - 14 \quad (a)$$

- (b) A function g is given by $g(x) = x^2 - 3x + 18$.

- (i) Find the equation of the tangent at the point on the graph of g where the gradient is 0.

$$y = mx + c$$

$$m = \text{gradient} = 0$$

$$g'(x) = 2x - 3$$

$$\text{when } g'(x) = 0$$

$$2x - 3 = 0$$

$$2x = 3$$

$$x = \frac{3}{2}$$

$$\text{gradient} = 0 \text{ when } x = \frac{3}{2}$$

$$g\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right) + 18 = 15.75$$

$$\text{so eqn when grad} = 0 \text{ is } y = 15.75 \quad (m)$$

- (ii) In relation to the graph, fully describe the point where this tangent meets the function.

$g(x)$ is a positive parabola



so grad = 0 at minimum.

Therefore the pt $(1.5, 15.75)$

is the minimum point of the curve. (a)

- (c) A skateboard park has a mound that is h metres high at the point where the horizontal distance, from a fixed point P, is x metres.

The mound can be modelled by

$$h = -0.5x^2 + 3x - 1.5$$



- (i) What is the maximum height of the mound? *max where $\frac{dh}{dx} = 0$.*

a) $\frac{dh}{dx} = -x + 3$
 $-x + 3 = 0$
 $3 = x$
 max occurs when $x = 3$

m) $h = -0.5(3)^2 + 3(3) - 1.5$
 $h = 3\text{m}$

- (ii) A ramp up the side of the mound is a tangent to the mound.

The ramp can be modelled by the function

$$h = 0.5x - c \leftarrow \text{tgt grad} = 0.5$$



Use calculus to find the vertical distance below the top of the mound where the ramp will meet the mound.

Ignore the thickness of the ramp.

- need to find x -value corresponding to a grad of 0.5
- then find h
- then compare with 3m.

a) $\frac{dh}{dx} = -x + 3$
 $-x + 3 = 0.5$
 $3 = 0.5 + x$
 $x = 2.5$
 when $x = 2.5$

m) $h = -0.5(2.5)^2 + 3(2.5) - 1.5$
 $h = 2.875\text{m}$

e) $3 - 2.875 = 0.125$
 so the ramp meets the mound 0.125m below the top of the mound.

- (iii) The height h metres of a skateboard path at a horizontal distance r metres from another point Q, can be modelled by the function

$$h = \frac{r^3}{3} - 2r^2 + 3r \quad (0.15 < r < 3.5)$$



There is a height regulation that requires no part of the skateboard path to be more than 3 m above the ground.

Will need to show if pts min/max

Fully describe this curve including its turning points, and state whether or not the skateboard path complies with the height regulation.

You must show calculus in answering this question.

- find min/max pts $\rightarrow \frac{dh}{dr} = 0$
- Show min/max
- find max height to compare with 3m.

$$\frac{dh}{dr} = r^2 - 4r + 3$$

$$r^2 - 4r + 3 = 0$$

$$(r-3)(r-1) = 0$$

$$r = 3 \text{ or } r = 1$$

$$\text{tp at } r=1 \text{ \& } r=3$$

nature of tp investigation.

r	0	1	2	3	4
$\frac{dh}{dr}$	3	0	-1	0	3
inc/dec	/	-	\	-	/

when

$$r=0 \quad \frac{dh}{dr} = 3 \rightarrow +ve \quad /$$

$$r=2 \quad \frac{dh}{dr} = -1 \rightarrow -ve \quad \backslash$$

$$r=4 \quad \frac{dh}{dr} = 3 \rightarrow +ve \quad /$$

so $r=1$ is a maximum

and $r=3$ is a minimum

could also have

- sketched carefully $h =$ cubic & discussed shape
- sketched $\frac{dh}{dr} =$ parabola & discussed where $\frac{dh}{dr} > 0$ & $\frac{dh}{dr} < 0$.

when $r=1$

$$h = \frac{1}{3} - 2 + 3$$

$$h = \frac{4}{3} \text{ m}$$

when $r=3$

$$h = \frac{27}{3} - 18 + 9$$

$$h = 0$$

the height of the ramp at its maximum is only $\frac{4}{3}$ m, so the ramp is well within the regulations.

NOTE: only reason to find $r=3$ is the marking schedule.



QUESTION THREE

- (a) The velocity $v \text{ m s}^{-1}$ of an object t seconds after it passes a fixed point can be modelled by the function

$$v(t) = 4t^3 - t^2 + 2t$$

velocity speed $\xrightarrow{\text{diff}}$ acc

Find the equation for the acceleration of the object.

$$a(t) = v'(t) = 12t^2 - 2t + 2$$

(a)

- (b) Find the equation of the tangent to the curve $f(x) = x^3 - 2x^2 + x$ at the point (2,2) on the curve.

tgt $\rightarrow$ need gradient at pt + pt
 $y = mx + c$

$$f'(x) = 3x^2 - 4x + 1$$

when $x = 2$ gradient:

$$f'(2) = 3(2)^2 - 4(2) + 1$$

$$f'(2) = 12 - 8 + 1 = 5$$

$$m = 5$$

$$y = 5x + c$$

substitute (2,2) to find c

$$2 = 5(2) + c$$

$$2 = 10 + c$$

$$c = -8$$

$$y = 5x - 8$$

(m)

- (c) In an area surrounding a farming airstrip there is a height restriction for fireworks of 50 m. The height h metres above the ground reached by a firework t seconds after it is fired, can be modelled by the function

$$h = 20t - 5t^2$$

Will the firework break the 50 m limit?

Use calculus methods to justify your answer.



Find height at maximum to see
 $\rightarrow$ maximum when $\frac{dh}{dt} = 0$

$$\frac{dh}{dt} = 20 - 10t$$

maximum:

$$20 - 10t = 0$$

$$20 = 10t$$

$$t = 2$$

maximum occurs at $t = 2 \text{ sec}$.

height at $t = 2 \text{ sec}$

$$h = 20(2) - 5(2)^2 = 20 \text{ m}$$

so firework will not break height restriction

max height + conclusion = (m)

- (d) For a function $y = -ax^2 + bx + c$,
 a , b , and c are positive numbers and $b = 2a$.

On the grid below, sketch the gradient function.

Show the value of all intercepts. The y' -intercept should be given in terms of b .

$$\frac{dy}{dx} = -2ax + b$$

$$b = 2a$$

(a) $= -2ax + 2a$

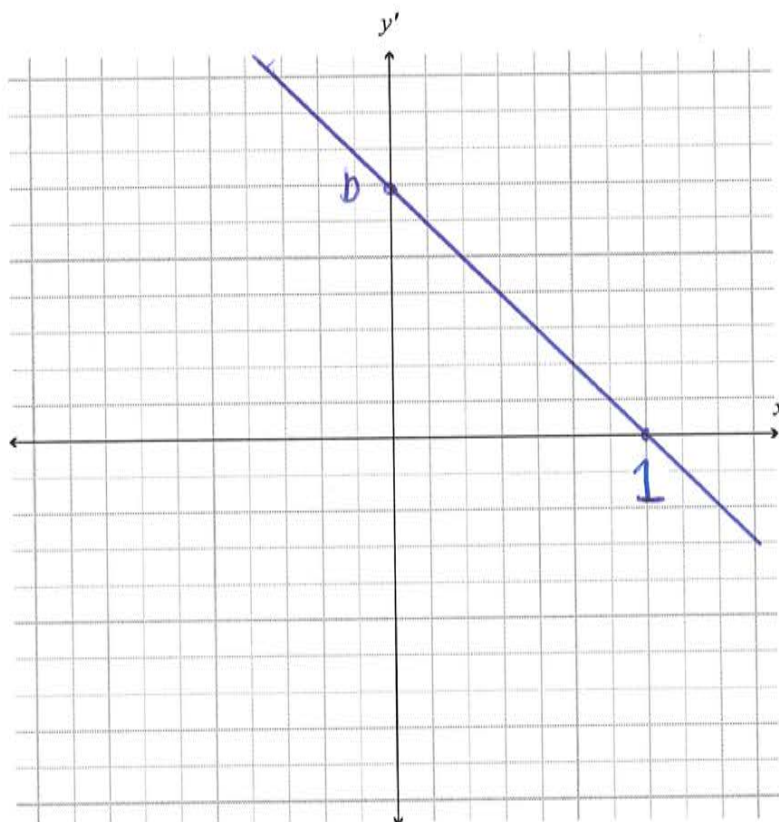
$$a = \frac{b}{2}$$

or

(a) $\frac{dy}{dx} = -bx + b$

$$\text{gradient} = -b$$

$$y\text{-int} = b$$



one intercept
correct = (m)

graph correct
 $y\text{-int} = b$
 $x\text{-int} = 1$ (c)

If you need
to redraw this
graph, use the
grid on page 10.

- (e) $y = (x-3) \times 2$
 y is the value of x after 3 has been subtracted and then the answer doubled, and x is between -0.5 and 3 .

Find the maximum and minimum values of the product of x^2y .

Justify your answer.

$$y = 2(x-3)$$

$$P = x^2y = 2x^2(x-3)$$

$$P = 2x^3 - 6x^2$$

P is a positive cubic
 so looks like .

$$P'(x) = 6x^2 - 12x$$

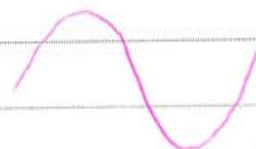
min/max when $P'(x) = 0$

$$6x^2 - 12x = 0$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x = 0 \text{ or } x = 2.$$



Smallest ^{stat pt} x -value corresponds
 to maximum ($x=0$)

Largest ^{stat pt} x -value corresponds
 to minimum ($x=2$)

when $x = 0$

$$P = 2(0)^3 - 6(0)$$

$$P = 0$$

maximum

when $x = 2$

$$P = 2(2)^3 - 6(2)^2$$

$$P = -8$$

minimum.